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I confirm that I understand my coursework needs to be submitted online via MST under the relevant module page before the deadline for my assignment to be accepted and marked. I am fully aware that late submissions will be treated as non-submission and a mark of zero will be awarded.

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Abstract

Cryptography is an essential tool for ensuring Confidentiality, Integrity, and Secure Communication throughout today's digital environments. In this course, you will learn about the basic concepts of Cryptography and see how Encryption Techniques have developed over time from Classical Ciphers to Sophisticated Cryptographic Methods. Using the Traditional Caesar Cipher as your foundation, this coursework has led to the creation of a new Cryptographic Algorithm called the Shital Cipher.

The Shital Cipher is unique because it adds many Layers of Encryption, such as Reverse Alphabet Substitution, Caesar Shifting, ASCII and Binary Conversion, XOR operations, and Permutation Techniques. These added Layers provide greater complexity in the encryption process, which increases Resistance Against Cryptanalysis. A number of different test cases were designed to assess the capabilities of the Shital Cipher, including tests that utilized Sensitive Data.

The Ultimate Outcome of these test cases reveals that the Shital Cipher functions as expected by Encrypting and Decrypting Data and generates Unpredictable Ciphertext. Additionally, this course will provide you with an understanding of how to apply Principles of Designing Cryptographic Systems in order to Achieve Layered Security for Digital Information with Cyber Security.

INTRODUCTION

Figure 1: Figure of History of Cryptography.

Throughout history, information has been hidden from spies through various techniques from time to time to have secure communication in world wars, military operations, etc. The field of secret writing is composed of two main branches Steganography and Cryptography Steganography is a secret writing or information-hiding technique derived from Greek and is defined as “cover writing”. Steganography aims to hide messages and transmit them to various channels such as text, images, and audio through a communication channel to protect their privacy from an eavesdropper. It is further classified into linguistic steganography and technical steganography. In linguistic steganography, messages are hidden in natural language, while technical steganography is a carrier rather than a text, which is used to transmit secret messages such as microdots and invisible links. Cryptography is another secret writing technique, where the sender and receiver share messages secretly through encryption and decryption methods using the keys. It is further classified into substitution and transposition ciphers. Each category varies with the use of letters or blocks in the encryption and decryption process (Tunio, 2024).

Aims And Objective

Aim

The aim of this coursework is to study the fundamental principles of cryptography and design a new cryptographic algorithm to secure data from unauthorized access. It also aims to enhance data confidentiality by applying mathematical and logical techniques and demonstrate the practical application of cryptographic system design in cyber security.

Objective

- To Investigate fundamental ideas of Cybersecurity with a focus on the relevance of Cryptography.
- To Conduct research into the growth and development of Cryptography through time.
- To Compare and contrast Symmetric Encryption and Asymmetric Encryption Algorithms.
- To Select from an already-existing Cryptography as the basis for creating a new Cipher.
- To Develop a new Cipher utilizing Logical and Mathematic Operations.
- To Create the processes for both Encrypting and Decrypting.
- To Construct a series of stepwise Algorithms and Flowchart for the proposed Cipher.
- To Prove that the Cybersecurity system you have designed meets the requirement of protecting Data Confidentiality.

History of Cryptography

Figure 2: Figures of History of Cryptography.

In 1957, Mary Queen of Scots utilized the science of cryptography in her plot to execute her cousin Queen Elizabeth I, with the help of her fellow companions who shared her intention to remove the Queen from the throne and replace her with Mary. The cryptography method of that time involved using papers and pens. Mary used to conceal her messages in the letters with encrypted text known as ‘Cipher text’. She then sent the letters through a person secretly working for the Queen. However, the insecure channel made it easy for her plans to be detected, as the Queen’s codebreaker, an expert cryptanalyst, intercepted and encrypted all of Mary’s and her companion’s letters, which were then used as evidence against them. The cryptanalyst went even further to add forged text in those letters to implicate all conspirators in the plan (Singh, 2009).

1930s Encryption Vs 1970s Encryption

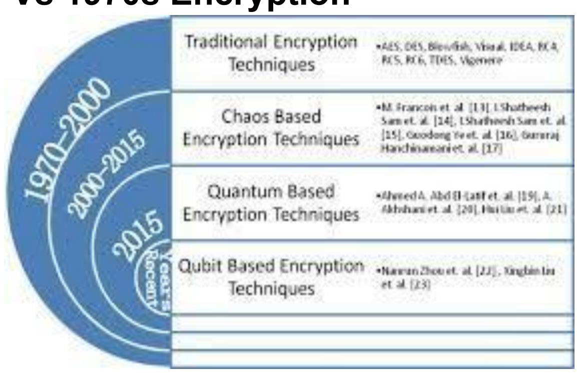

Figure 3: Figure of Evolution of the encryption techniques.

In the early days of encryption, one of the most significant historical examples of the ramifications of poor encryption is the situation surrounding "Mary, Queen of Scots". Mary Queen of Scots provides a well-documented example of what can happen when weak encryption is exploited or compromised in early examples of cryptography. During this time, Mary and Babington used a very weak encryption method referred to as a simple substitution cipher in these correspondences between them and therefore both parties assumed their encryption was fully capable of being secure from decryption efforts by any other parties. Unfortunately, both Mary and Babington were mistaken in this assumption, as the method of encryption that they used would have been so primitive it would be easily broken by contemporary cryptographers; as a consequence, both Mary and Babington were found guilty and sentenced to execution due to their poor judgment regarding the security of their method of encryption.

The defeat of both parties lent credence to the push for increased security in the field of cryptography resulting in the development of the Caesar Cipher, Vigenère Cipher, and many varieties of complex polyalphabetic Ciphers, making it more difficult for cryptates to successfully decode text encrypted using earlier techniques. Early examples of encryptions were predominantly paper based systems but later advancements in encryption technology represented major strides toward the science of cryptography. Historically it was believed that if a particular encryption design had been designed with secrecy maintained this, therefore meant that the design for the system would remain secure from the efforts of others until the system was discovered or otherwise leaked to other Third Parties (Paar, 2009).

Modern Cryptography

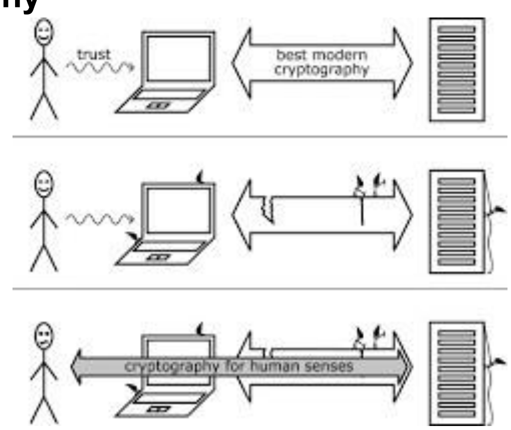

Figure 4: Figures of Modern Cryptography.

In the current technological era of cryptography, the emergence of computers and the internet revolutionized the industry, using the ideas of Claude Shannon. Shannon was the first person to use the term "cryptography," as he developed many of the original methods and techniques that serve to secure and authenticate information over time through his work at Bell Labs. Shannon's ideas are still important to many researchers today.

Most of the early computer-based ciphers were similar to mechanical devices, such as the Enigma machine. However, they were designed to be quicker and more efficient than previous machines. An example of this would be the IBM Lucifer Cipher. The Lucifer Cipher developed by IBM was the basis for the Data Encryption Standard (DES), which was adopted by the U.S. government as its official encryption standard. DES established the concept of "block ciphers," but ultimately, due to the short length of the key used in DES, it was compromised and this resulted in the development and adoption of the Advanced Encryption Standard (AES). The AES uses a number of different key sizes ranging from 128 bits up to 256 bits.

To solve the key distribution problem, Rivest, Shamir, and Adleman created public key cryptography (PKC), referred to as RSA. With the invention of RSA, secure communications could be achieved through the use of a Public Key and a Private Key, thus forming the basis of digital signatures. However, with the advancements of quantum computing, RSA may be at risk due to the ability of quantum computers to efficiently factor extremely large prime numbers, thereby potentially causing a breach of the RSA system (Lewis, 2019).

Overview of Symmetric and Asymmetric Encryption

Symmetric Encryption

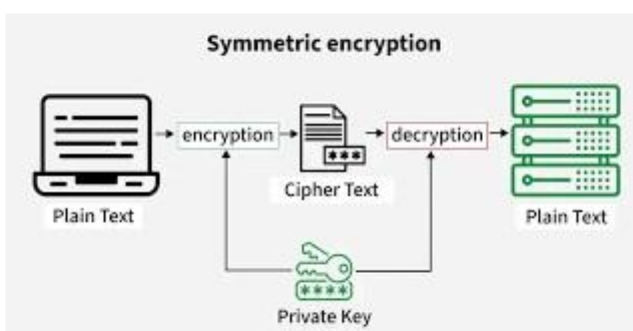

Figure 5: Figure of Symmetric encryption.

Two communicating parties share a single secret encryption key (symmetric encryption) which is used both ways (i.e., to encrypt and decrypt). This means that if two people need to communicate securely, they both have to obtain the same secret encryption key in order to communicate. A symmetric encryption method will work very quickly to encrypt large amounts of information compared to asymmetric encryption methods; therefore, symmetric encryption methods are much faster in comparison to asymmetric encryption methods. The hard part when it comes to symmetric encryption is how to securely distribute the secret encryption key from one party to another. If any outside party is able to intercept the secret encryption key while it is being transmitted, the original file will no longer be secure, and subsequently, the original file will no longer have its confidentiality maintained (willim, 2025).

Advantages of Symmetric Encryption

- Faster processing speed compared to asymmetric encryption.
- Less computationally intensive, making it efficient for large data.
- Simpler algorithms lead to easier implementation.
- Ideal for real-time data encryption.
- Key management and distribution can be challenging

Asymmetric Encryption

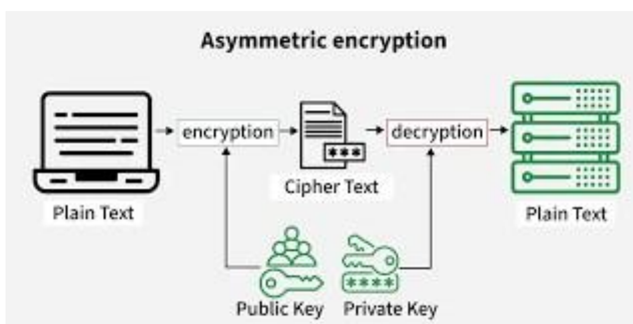

Figure 6: Figures of Asymmetric Encryption.

Asymmetric cryptography is based on the use of two separate but related keys (a public key that may be freely distributed and a private key that must be kept private) to protect the confidentiality of the communications between two parties. Anyone can use a public key to encrypt their message, but a recipient must have the matching private key to decrypt it. The advantage of using an asymmetric method of cryptography has resulted in improved security, especially related to key-sharing processes. However, the asynchronous nature of cryptography has produced significant performance penalty's when encrypting and decrypting data; therefore, it is less efficient for encrypting large amounts of data than symmetric cryptography.

Advantages of Asymmetric Encryption

- Enhanced security for key exchanges without sharing private keys.
- Enables digital signatures for data integrity and authenticity.
- Suitable for environments where users do not have prior relationships.
- Supports secure communications over open networks.
- Balances security with the need for data confidentiality.

Use Cases for Symmetric and Asymmetric Encryption

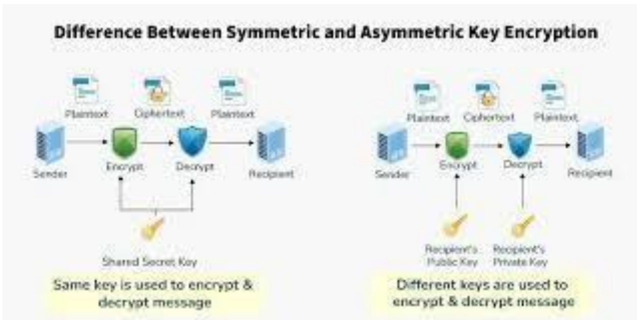

Figure 7: Figure of Symmetric and Asymmetric Encryption.

- Symmetric encryption is ideal for internal data transfer and storage.
- Commonly used in VPNs (Virtual Private Networks) for secure connections.
- Asymmetric encryption is essential for secure email protocols like PGP (Pretty Good Privacy).
- Widely used in SSL/TLS protocols for establishing secure web connections.
- Digital signatures are employed in software distribution to verify integrity.

Background of the Selecting Cryptography Algorithm

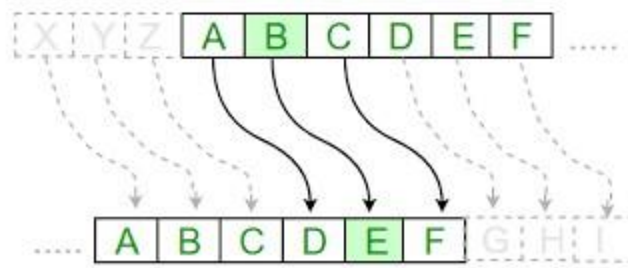

Figure 8: Figures of Caeser Cipher.

The Cryptography algorithm selected for this coursework is the "Caesar Cipher". The Caesar cipher is one of the oldest known cryptographic methods, named after Julius Caesar, who reportedly used it to keep his military messages secret. It works by systematically replacing each letter in a message with another letter a fixed number of positions away in the alphabet, a simple but clever way to keep intercepted orders useless without the right key (Owolabi, 2025).

The Caesar cipher, also known as Caesar's cipher and Caesar's code, consists of a fixedlength alphabetic shift on each letter in the plain text to create a ciphered or encoded text.
In a shift of 3, the letter A would become D, B would become E, etc. The name originates from the Roman leader and military strategist, Julius Caesar, who utilized the system for his military communications with generals during his rule.

Although the Caesar cipher is an encrypted alphabetic substitution technique, it is commonly used as a basic building block of more complex encryption schemes, such as the Vigenère cipher, which is used for encrypting and decrypting text. The ROT13 system is the most current application of the Caesar cipher as it is still used in contemporary applications. Like all of the other single alphabet substitution techniques, the Caesar cipher is relatively simple to decode, and in today's world, it provides no real security.

Used of Caesar Cipher

The transformation can be represented by aligning two alphabets; the cipher alphabet is the plain alphabet rotated left or right by some number of positions. For instance, here is a Caesar cipher using a left rotation of three places (the shift parameter, here 3, is used as the key):

Plain: ABCDEFGHIJKLMNOPQRSTUVWXYZ

Cipher: DEFGHIJKLMNOPQRSTUVWXYZABC

When encrypting, a person looks up each letter of the message in the "plain" line and writes down the corresponding letter in the "cipher" line. Deciphering is done in reverse.

Cipher text: WKH TXLFN EURZQ IRA MXPSV RYHU WKH ODCB GRJ 2

Plaintext: the quick brown fox jumps over the lazy dog

The encryption can also be represented using modular arithmetic by first transforming the letters into numbers, according to the scheme, A = 0, B = 1..., Z = 25.[1] Encryption of a letter x by a shift n can be described mathematically as,

$E_n(x) = (x + n) \bmod 26$

Decryption is performed similarly,

$D_n(x) = (x - n) \bmod 26$

There are different definitions for the modulo operation. In the above, the result is in the range 0...25. I.e., if x + n or x - n are not in the range 0...25, we have to subtract or add 26. The replacement remains the same throughout the message, so the cipher is classed as a type of monoalphabetic substitution, as opposed to polyalphabetic substitution.

Advantage of Caeser Cipher

- Easy to implement and use thus, making suitable for beginners to learn about encryption.
- Can be physically implemented, such as with a set of rotating disks or a set of cards, known as a scytale, which can be useful in certain situations.
- Requires only a small set of pre-shared information.
- Can be modified easily to create a more secure variant, such as by using a multiple shift values or keywords.

Disadvantage of Caeser Cipher

- It is not secure against modern decryption methods.

- Vulnerable to known-plaintext attacks, where an attacker has access to both the encrypted and unencrypted versions of the same messages.
- The small number of possible keys means that an attacker can easily try all possible keys until the correct one is found, making it vulnerable to a brute force attack.
- It is not suitable for long text encryption as it would be easy to crack.
- It is not suitable for secure communication as it is easily broken.
- Does not provide confidentiality, integrity, and authenticity in a message.

Development of a new Cryptographic Algorithm

This coursework presents several improvements to the traditional Caesar Cipher to increase its security. Each of these improvements provides another level of sophistication for encryption and decryption. In addition, these changes help address the vulnerabilities of the original Caesar Cipher. The end result is a completely new encryption algorithm that has a greater ability to withstand different methods of cryptanalysis such as brute force searching or frequency analysis.

The new encryption algorithm created from these improvements is referred to as the Shital Cipher. The Shital Cipher is a symmetric-style key-based cipher built from the original Caesar Cipher; it has been improved with an added layer of complexity and the use of numbers and binary representations, as well as other types of math and logic operations.

First Modification: Reversed Alphabet Substitution

The first modification applied to the Caesar Cipher involves an **alphabetic variation**, where the standard alphabetical order is rearranged in reverse. Instead of using the conventional alphabet sequence (A–Z), the Shital Cipher employs a reversed alphabet (Z–A). This modification increases unpredictability in the substitution process and makes the cipher more resistant to basic frequency analysis.

Original Alphabet (Caeser Cipher)

The original arrangement of the alphabet used in Caesar cipher is given below:

Table 1: Table of Original Alphabet (Caeser Cipher).

A	B	C	D	E
F	G	H	I	J
K	L	M	N	O
P	Q	R	S	T
U	V	W	X	Y
Z				

Modified Alphabet (Shital Cipher)

The modified arrangement of the alphabet in Shital Cipher to Reversed of the algorithm is given below:

Table 2: Table of Modified Alphabet(Shital cipher).

Z	Y	X	W	V
U	T	S	R	Q
P	O	N	M	L
K	J	I	H	G
F	E	D	C	B
A				

Second Modification: ASCII and Binary Transformation (Mathematical Form)

The second modification of the Shital Cipher enhances security by converting the substituted characters into their ASCII values and then into binary form. This transformation shifts the encryption process from a simple alphabetical level to a numerical and bit-level representation. By operating in the binary domain, the cipher significantly increases complexity and reduces the effectiveness of classical cryptanalysis techniques such as frequency analysis.

- $P = \{P_0, P_1, \dots, P_n\}$ be the plaintext
- K be the secret key (assume K = 2)
- $A(P_i)$ be the alphabetical index using the reversed alphabet
- C_i be the intermediate encryption character

Step 1: Modification Caeser Encryption

The substitution rule is $C_i = (\mathrm{A}(P_i) + K) \bmod 26$

Plaintext

FOOTBALL

Alphabetical Mapping (Reversed Alphabet)

Table 3: Table of Alphabetical Mapping (Reversed Alphabet).

Letter	Index
F	21
O	11
O	11
T	6
B	24
A	25
L	14
L	14

Applying Key (K = 2)

$$C_i = (Index + 2) mod\ 26$$

Table 4: Table of Applying Key (K=2).

Letter	Result Index	Cipher Letter
F	23	D
O	13	M
O	13	M
T	8	S
B	0	Z
A	1	Y
L	16	J

L	16	J

Step 2: ASCII Conversion

$D_i = \text{ASCII}\ (C_i)$

Table 5: Table of ASCII Conversion.

Cipher Letter	ASCII
D	68
M	77
M	77
S	83
Z	90
Y	89
J	74
J	74

Step 3: Binary Representation

$B_i = \text{Binary}\ (D_i)$

Table 6: Table of Binary Representation.

ASCII	Binary
68	1000100
77	1001101
77	1001101
83	1010011
90	1011010
89	1011001

74	1001010
74	1001010

Third Modification: XOR and Permutation (Mathematical Form)

- $S = 1100110$ be the fixed XOR sequence
- $P_7 = (7,2,5,1,3,6,4)$ be the permutation Vector

Step 1: XOR Operation

$$X_i = B_i \text{ XOR } S$$

Example for First Letter

1000100 XOR 1100110 = 0100010

Applying XOR to all values:

Table 7: Table of XOR Operation.

Binary	XOR Result
1000100	0100010
1001101	0101011
1001101	0101011
1010011	0110101
1011010	0111100
1011001	0111111
1001010	0101100
1001010	0101100

Step 2: Permutation

Permutation rule: $Y_i(j) = X_i(P_7(j))$

Applying permutation to all the first XOR output:

0100010 = 0010100

Applying permutation to all the values:

Table 8: Table of Applying Permutation of all the values.

XOR Output	After Permutation
0100010	0010100
0101011	1010100
0101011	1010100
0110101	1010011
0111100	0011110
0111111	1110111
0101100	0011100
0101100	0011100

Step 3: ASCII Reconstruction

$C_i' = ASCII^{-1}(Decimal\ (Y_i))$

Table 9: Table of ASCII Reconstruction.

Binary	Decimal	Character
0010100	20	DC4
1010100	84	T
1010100	84	T
1010011	83	S
0011110	30	RS
1110111	119	W
0011100	28	FS
0011100	28	FS

Then the Final Ciphertext is

T T S w

Algorithm of Encryption and Decryption

Encryption

Step 1: Read the plaintext message P.

Step 2: Select the secret key K.

Step 3: Replace the standard alphabet with the reversed alphabet (Z-A).

Step 4: for each character Pi in the plaintext:

- Map Pi to its alphabetical index using the reversed alphabet.
- Computing the modified Caeser shift:

$$C_i = ((P_i) + K) \bmod 26$$

- Convert Ci into its ASCII values.
- Convert the ASCII value into a 7-bit binary representation.
- Apply the XOR operation using the binary key S.
- Applying the permutation sequence P7 to the XOR result.

- Convert the permuted binary values back to decimal.
- Map the decimal value to a character using the ASCII table.

Step 5: Concatenate all resulting character to form the final ciphertext C. Step

6: Stop

Decryption

Step 1: Read the ciphertext message C.

Step 2: Use the same reversed alphabet, XOR key S, and permutation sequences P7.

Step 3: For each character Ci in the ciphertext:

- Convert Ci into its ASCII value.
- Convert the ASCII value into a 7-bit Binary representation.
- Apply the reverse permutation of P7.
- Apply the XOR operation using the same binary key S.
- Convert the resulting binary value into decimal.
- Convert the decimal values to its corresponding character using the ASCII table.
- Apply the inverse Caeser shift:

$$Pi = (Ci - K) \bmod 26$$

- Map the result back to the original letter using the reversed alphabet.

Step 4: Concatenate all the characters to reversed the original plaintext P.

Step 5: Stop.

Flowchart of the Encryption and Decryption

Encryption

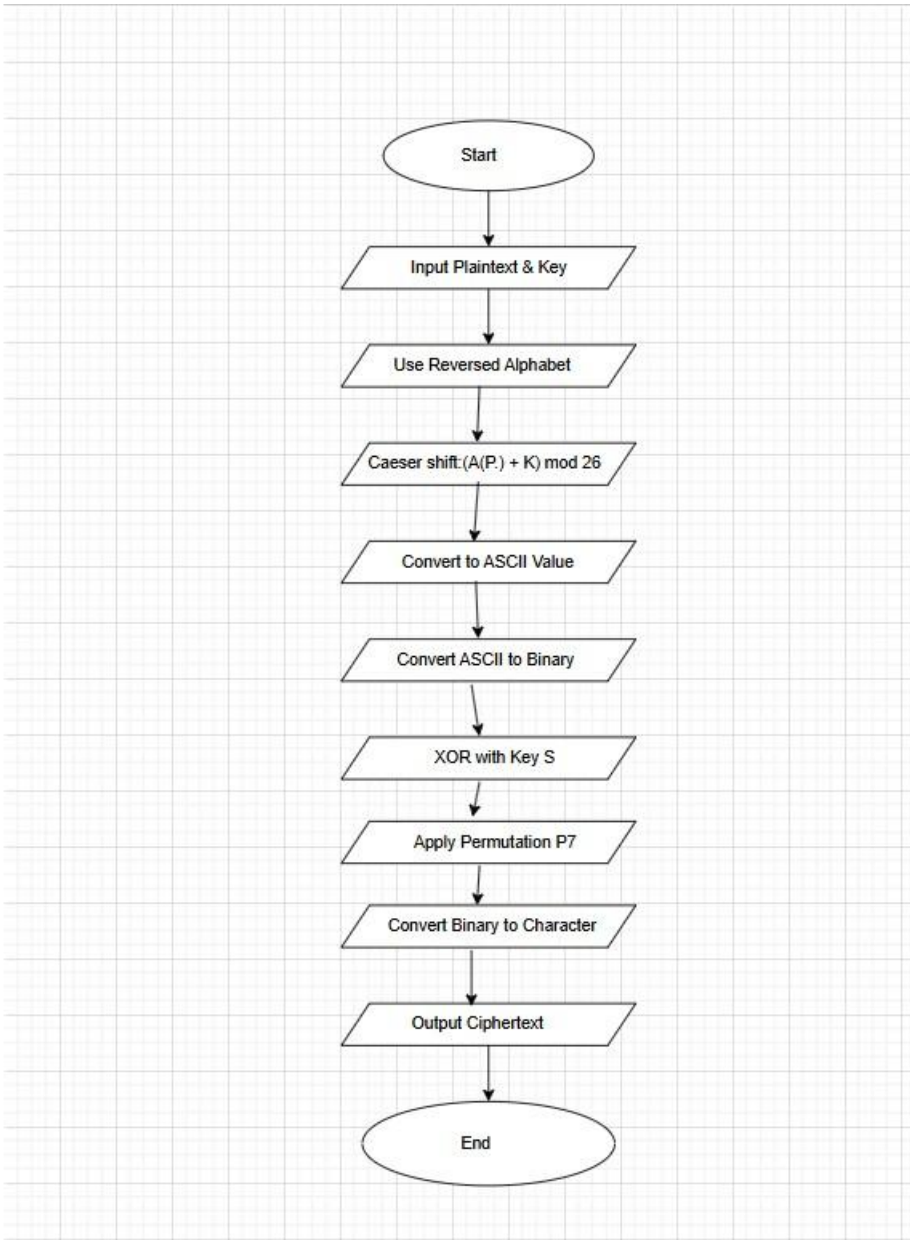

Figure 9: Figures of Flowchart of the Encryption.

Decryption

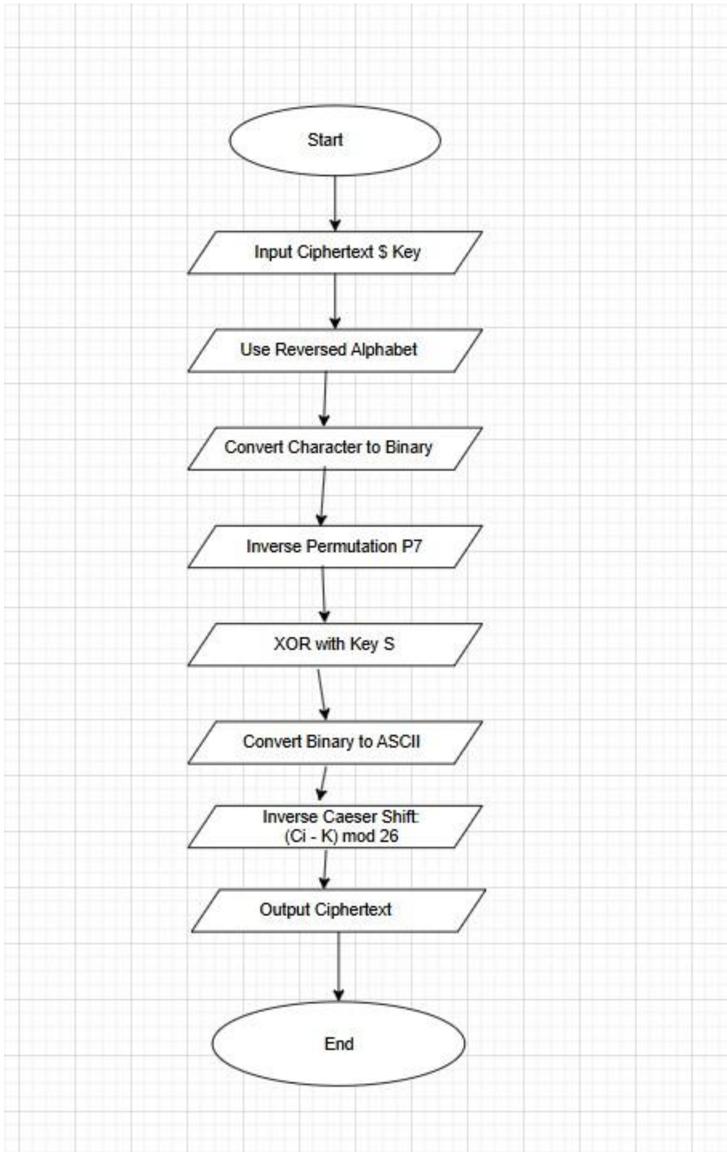

Figure 10: Figures of Flowchart of the Decryption.

Testing

During the testing phase, the accuracy, performance, and dependability of the new Shital Cipher were verified. Every test case represents a function of the cipher: its ability to encrypt correctly, decrypt correctly, depend on a unique key, and resist a wide spectrum of basic attacks. In order to gain the most accurate results possible about how well the algorithm works, multiple test cases were run with both dissimilar inputs as well as varying key values.

Test case 1

Test Case 1 was designed to verify the correct encryption and decryption of a longer plaintext using the Shital Cipher. The test applies all stages of the algorithm, including reverse alphabet substitution, Caesar shifting, ASCII and binary conversion, XOR operation, and permutation. Successful recovery of the original plaintext confirms the correctness and reliability of the encryption and decryption processes for longer character sequences.

Encryption Process

- Plaintext = FOOTBALL
- Key (K) = 3

Reversed Alphabet Substitution

Table 10: Table of Reversed Alphabet Substitution.

Plaintext	Alphabet Position	Reversed Alphabet
F	5	U
O	14	L
O	14	L
T	19	G
B	1	Y
A	0	Z
L	11	O
L	11	O

Ceaser Shifting (Key = 3)

Formula used: C = (P+K) mod 26

Table 11: Table of Ceaser Shifting with Key = 3.

Reversed Letter	Position	Shifted Position	Shifted Letter
U	20	23	X
L	11	14	O
L	11	14	O
G	6	9	J
Y	24	1	B
Z	25	2	C
O	14	17	R
O	14	17	R

ASCII Conversion

Table 12: Table of ASCII Conversion.

Shifted Letter	ASCII Value
X	88
O	79
O	79
J	74
B	66
C	67
R	82
R	82

Binary Conversion (7-bit)

Table 13: Table of Binary Conversion.

ASCII Value	Binary Number
88	1011000
79	1001111
79	1001111
74	1001010
66	1000010
67	1000011
82	1010010
82	1010010

XOR Operation

XOR Key (S1): 1100110

Table 14: Table of XOR Operation.

Binary Input	XOR Key	Result
1011000	1100110	0111110
1001111	1100110	0101001
1001111	1100110	0101001
1001010	1100110	0101100

1000010	1100110	0100100
1000011	1100110	0110100
1010010	1100110	0110100
1010010	1100110	0110100

Permutation

Permutation Rule: P7 = (7,2,5,1,3,6,4)

Table 15: Table of Permutation.

XOR Output	After Permutation
0111110	1111010
0101001	0101011
0101001	0101011
0101100	0100011
0100100	0010010
0100101	0101010
0110100	1101010
0110100	1101010

Final Ciphertext Generation

Table 16: Tables of Final Ciphertext Generation.

Permuted Binary	ASCII	Cipher Character
1111010	122	z
0101011	43	+
0101011	43	+
0100011	35	#
0010010	18	@
0101010	42	&
1101010	106	j
1101010	106	j

The final Ciphertext of

FOOTBALL = z++#@&jj

Decryption Process

Input

- **Ciphertext:** z++#@&jj • **Key:** 3

ASCII Conversion

Table 17: Table of ASCII Conversion.

Cipher Text	ASCII
z	122
+	43
+	43
#	35
@	18
&	42
j	106
j	106

Binary Conversion

Table 18: Table of Binary Conversion.

ASCII	Binary
122	1111010
43	0101011
43	0101011
35	0100011
18	0010010
42	0101010
106	1101010
106	1101010

Inverse Permutation

Table 19: Table of Inverse Permutation.

Permuted Binary	After Inverse Permutation
1111010	0111110
0101011	0101001
0101011	0101001
0100011	0101100
0010010	0100100
0101010	0100101
1101010	0110100
1101010	0110100

XOR Reversal

Table 20: Table of XOR Reversal.

XOR Result	XOR Key	Original Binary
0111110	1100110	1011000
0101001	1100110	1001111
0101001	1100110	1001111
0101100	1100110	1001010
0100100	1100110	1000010
0100101	1100110	1000011
0110100	1100110	1010010
0110100	1100110	1010010

Binary to ASCII

Table 21: Table of Binary to ASCII Conversion.

Binary Result	ASCII
1011000	88
1001111	79
1001111	79
1001010	74
1000010	66
1000011	67

1010010	82
1010010	82

Reverse Caesar Shift

P = (C - K) mod 26

Table 22: Table of Reverse Caesar Shift.

Shifted	After Shift
X	U
O	L
O	L
J	G
B	Y
C	Z
R	O
R	O

Reversed Alphabet Mapping

Table 23: Table of Reversed Alphabet Mapping.

Reversed Text	Plain Text
U	F
L	O
L	O
G	T
Y	B
Z	A
O	L
O	L

The final Recovered Plaintext of

z++#@&jj = FOOTBALL

Here is the Result of the test 1.

Table 24: Table of Result of the test 1.

Objective	To verify correct encryption and decryption of a longer plaintext using the Shital Cipher.
Action	The plaintext “FOOTBALL” is encrypted using the Shital Cipher, which applies reverse alphabet substitution, Caesar shifting with key 3, ASCII conversion, binary conversion, XOR operation, and permutation. The resulting ciphertext is then decrypted by applying inverse permutation, XOR decryption, reverse Caesar shifting, and reverse alphabet mapping.
Conclusion	The original plaintext is successfully reconstructed, confirming that the Shital Cipher functions correctly for longer character sequences.
Result	Pass

Test case 2

Encryption Process

- Plaintext = CRYPTO
- Key = 3
- XOR Key = 1100110
- Permutation (P7) = (7,2,5,1,3,6,4)

Table 25: Table of test 2 Encryption Process.

Plaintext	Reverse text	Shifted (K = 3) C = (p + k) mod 26	ASCII Value	Binary	XOR Operation	XOR Output	After Permutation	ASCII Value	Cipher text
C	X	A	65	1000001	1100110	0100111	1110100	116	t
R	I	L	76	1001100	1100110	0101010	0101011	43	+
Y	B	E	69	1000101	1100110	0100011	1100010	98	b
P	K	N	78	1001110	1100110	0101000	0100011	35	#
T	G	J	74	1001010	1100110	0101100	0100011	35	#
O	I	O	79	1001111	1100110	0101001	0101011	43	+

The Final Cipher text of

CRYPTO = t + b # # +

Decryption Process

- Cipher text = t + b # # +
- Key = 3

Table 26: Table of Decryption process of Test 2.

Cipher text	ASCII Value	Binary	After inverse	XOR Key	XOR Output	ASCII Value	Letter	Reverse Caeser Shift (Key=3)	Plaintext
t	116	1110100	0100111	1100110	1000001	65	A	X	C
+	43	0101011	0101010	1100110	1001100	76	L	I	R
b	98	1100010	0100011	1100110	1000101	69	E	B	Y
#	35	0100011	0101000	1100110	1001110	78	N	K	P
#	35	0100011	0101100	1100110	1001010	74	J	G	T
+	43	0101011	0101001	1100110	1001111	97	O	L	O

The final recovered plaintext of

t + b # # + = CRYPTO

Here is the result of Test 2.

Table 27: Table of Result of the test 2.

Objective	To verify correct encryption and decryption of medium-length plaintext using Shital Cipher
Action	The plaintext CRYPTO is encrypted using reverse alphabet substitution, Caesar shift (K=3), ASCII and binary conversion, XOR operation, and permutation, then decrypted using inverse steps
Conclusion	The original plaintext CRYPTO is successfully reconstructed, confirming correct operation of the Shital Cipher
Result	Pass

Test Case 3

Test Case 3 was conducted to evaluate the key sensitivity and correctness of the Shital Cipher using a different plaintext and a new key value. The test confirms that changing the key results in a different ciphertext while still allowing accurate decryption. Successful recovery of the original plaintext demonstrates the reliability and key-dependent behaviour of the Shital Cipher.

Encryption Process

- Plaintext = NETWORK
- KEY (K) = 4
- XOR Key = 1100110
- Permutation (P7) = (7, 2, 5, 1, 3, 6, 4)

Table 28: Table of encryption process of test 3.

Plaintext	Reverse text	Shifted (K = 4) C = (p + k) mod 26	ASCII Value	Binary	XOR Operation	XOR Output	After Permutation	ASCII Value	Cipher text
N	M	Q	81	1010001	1100110	0110111	1111100	124	|
E	V	Z	90	1011010	1100110	0111100	0101111	47	/
T	G	K	75	1001011	1100110	0101101	1100011	99	c
W	D	H	72	1001000	1100110	0101110	0100111	39	‘
O	L	P	80	1010000	1100110	0110110	0111100	60	<
R	I	M	77	1001101	1100110	0101011	1101010	106	j
K	P	T	84	1010100	1100110	0110010	0100111	39	‘

The final Cipher text of

NETWORK = | / c ‘ < ‘ j

Decryption Process

- Cipher text = | / c ‘ < ‘ j
- Key (K) = 4

Table 29: Table of Decryption process of test 3.

<table>
<tr><th>Cipher text</th><th>ASCII Value</th><th>Binary</th><th>After inverse</th><th>XOR Key</th><th>XOR Output</th><th>ASCII Value</th><th>Letter</th><th>Reverse Caeser Shift (Key=3)</th><th>Plaintext</th></tr>
<tr><td>|</td><td>124</td><td>1111100</td><td>0110111</td><td>1100110</td><td>1010001</td><td>81</td><td>Q</td><td>M</td><td>N</td></tr>
<tr><td>/</td><td>47</td><td>0101111</td><td>0111100</td><td>1100110</td><td>1011010</td><td>90</td><td>Z</td><td>V</td><td>E</td></tr>
<tr><td>c</td><td>99</td><td>1100011</td><td>0101101</td><td>1100110</td><td>1001011</td><td>75</td><td>K</td><td>G</td><td>T</td></tr>
<tr><td>‘</td><td>39</td><td>0100111</td><td>0101110</td><td>1100110</td><td>1001000</td><td>72</td><td>H</td><td>D</td><td>W</td></tr>
<tr><td><</td><td>60</td><td>0111100</td><td>0110110</td><td>1100110</td><td>1010000</td><td>80</td><td>P</td><td>L</td><td>O</td></tr>
<tr><td>j</td><td>106</td><td>1101010</td><td>0101011</td><td>1100110</td><td>1001101</td><td>77</td><td>M</td><td>I</td><td>R</td></tr>
<tr><td>‘</td><td>39</td><td>0100111</td><td>0110010</td><td>1100110</td><td>1010100</td><td>84</td><td>T</td><td>P</td><td>K</td></tr>
</table>

The final recovered Plain test of

| / c ‘ < ‘ j = N E T W O R K

Here is the Result of Test 3

Table 30: Table of test result of the test 3.

Objective	To verify key sensitivity and correctness of the Shital Cipher using a different plaintext
Action	The plaintext NETWORK is encrypted using reverse alphabet substitution, Caesar shifting with key 4, ASCII and binary conversion, XOR operation, and permutation, then decrypted using inverse operations
Conclusion	The original plaintext NETWORK is successfully recovered, confirming correct functionality of the Shital Cipher
Result	Pass

Test Case 4

Test Case 4 evaluates the Shital Cipher using a new plaintext, a different key value, and a different XOR operation. The successful recovery of the original message confirms the adaptability of the cipher and demonstrates that modifying encryption parameters produces different ciphertext while maintaining correct decryption.

Encryption Process

- Plaintext = DATABASE
- Key (K) = 2
- XOR Key = 1011011
- Permutation (P7) = (7, 2, 5, 1, 3, 6, 4)

Table 31: Table of encryption process of test 4.

Plaintext	Reverse text	Shifted (K = 4) C = (p + k) mod 26	ASCII Value	Binary	XOR Operation	XOR Output	After Permutation	ASCII Value	Cipher text
D	W	Y	89	1011001	1011011	0000010	0001000	8	*
A	Z	B	66	1000010	1011011	0011001	1100010	98	b
T	G	I	73	1001001	1011011	0010010	0100010	34	“
A	Z	B	66	1000010	1011011	0011001	1100010	98	b
B	Y	A	65	1000001	1011011	0011010	0101010	42	*
A	Z	B	66	1000010	1011011	0011001	1100010	98	b
S	H	J	74	1001010	1011011	0010001	1000010	66	B
E	V	X	88	1011000	1011011	0000011	1100000	96	‘

The final Cipher text of

DATABASE = * b “ b * b B ‘

Decryption process

- Cipher text = * b “ b * b B ‘
- Key (K) = 2

Table 32: Table of decryption process of the test 4.

Cipher text	ASCII Value	Binary	After inverse	XOR Key	XOR Output	ASCII Value	Letter	Reverse Caeser Shift (Key=2)	Plaintext
*	8	0001000	0000010	1011011	1011001	89	Y	W	D
b	98	1100010	0011001	1011011	1000010	66	B	Z	A
“	34	0100010	0010010	1011011	1001001	73	I	G	T
b	98	1100010	0011001	1011011	1000010	66	B	Z	A
*	42	0101010	0011010	1011011	1000001	65	A	Y	B
b	98	1100010	0011001	1011011	1000010	66	B	Z	A
B	66	1000010	0010001	1011011	1001010	74	J	H	S
‘	96	1100000	0000011	1011011	1011000	88	X	V	E

The final recovered plaintext of

* b “ b * b B ‘ = D A T A B A S E

Here is the Result of test 4

Table 33: Table of test result of the test 4.

Objective	To verify encryption and decryption using a different key and XOR operation
Action	The plaintext DATABASE is encrypted using Shital Cipher with key 2 and XOR key 1011011, then decrypted using inverse operations
Conclusion	The original plaintext DATABASE is successfully recovered, confirming correctness and flexibility of the Shital Cipher
Result	Pass

Test Case 5

Test Case 5 focuses on encrypting sensitive information such as passwords. The multilayered approach of the Shital Cipher ensures that even highly confidential data is transformed into unreadable ciphertext while still allowing accurate recovery through decryption, validating its suitability for secure data protection.

Encryption Process

- Cipher Name = Shital Cipher
- Plaintext = PASSWORD
- Key (K) = 4
- XOR Key = 1100101
- Permutation (P7) = (4, 7, 2, 6, 1, 5, 3)

Table 34: Table of Encryption process of the test 5.

Plaintext	Reverse text	Shifted (K = 4) C = (p + k) mod 26	ASCII Value	Binary	XOR Operation	XOR Output	After Permutation	ASCII Value	Cipher text
P	K	O	79	1001111	1100101	0101010	1010010	82	R
A	Z	D	68	1000100	1100101	0100001	0001010	10	*
S	H	L	76	1001100	1100101	0101001	1000011	67	C
S	H	L	76	1001100	1100101	0101001	1000011	67	C
W	D	H	72	1001000	1100101	0101101	1100010	98	b
O	L	P	80	1010000	1100101	0110101	0101110	46	.
R	I	M	77	1001101	1100101	0101000	0000011	3	*
D	W	A	65	1000001	1100101	0100100	0100001	33	!

The final Cipher Text of

PASSWORD = R * C C b . * !

Decryption Process

- Cipher Text = R * C C b . * !
- Key (k) = 4

Table 35: Table of Decryption process of the test 5.

Cipher text	ASCII Value	Binary	After inverse	XOR Key	XOR Output	ASCII Value	Letter	Reverse Caeser Shift (Key=4)	Plaintext
R	82	1010010	0101010	1100101	1001111	79	O	K	P
*	10	0001010	0100001	1100101	1000100	68	D	Z	A
C	67	1000011	0101001	1100101	1001100	76	L	H	S
C	67	1000011	0101001	1100101	1001100	76	L	H	S
b	98	1100010	0101101	1100101	1001000	72	H	D	W
.	46	0101110	0110101	1100101	1010000	80	P	L	O
*	3	0000011	0101000	1100101	1001101	77	M	I	R
!	33	0100001	0100100	1100101	1000001	65	A	W	D

The final Recovered Plaintext of

R * C C b . * ! = PASSWORD

Here is the test result of test 5

Table 36: Table of test result of the test 5.

Objective	To evaluate the Shital Cipher using highly sensitive data
Action	The word PASSWORD is encrypted using a different key, XOR pattern, and permutation, then decrypted
Conclusion	The original sensitive data is successfully recovered, demonstrating that the Shital Cipher can securely protect confidential information
Result	Pass

Evaluation

This section evaluates the developed **Shital Cipher** by analysing its advantages, disadvantages, strengths, weaknesses, opportunities, threats, and possible application areas. The evaluation helps to understand the effectiveness, practicality, and limitations of the proposed cryptographic algorithm.

Advantages of the Shital Cipher

- The Shital Cipher improves the security of the traditional Caesar Cipher by introducing multiple layers of encryption.
- Reverse alphabet substitution reduces predictability and resists simple frequency analysis.
- Conversion to ASCII and binary adds mathematical complexity to the encryption process.
- XOR and permutation operations provide bit-level diffusion and confusion.
- The cipher is fully reversible when the correct key is used.
- It is simple to understand and implement for educational and academic purposes.

Disadvantages of the Shital Cipher

- The algorithm is computationally slower than classical Caesar Cipher due to multiple processing steps.
- It is not suitable for real-time large-scale data encryption.
- The cipher does not provide strong protection against modern cryptanalytic attacks.
- Key management is simple and may be vulnerable if the key is exposed.
- It is not resistant to brute-force attacks when compared to modern encryption standards like AES.

SWOT Analysis of the Shital Cipher

Strengths

- Multi-layer encryption approach increases security.
- Combines substitution, mathematical operations, and bitwise logic.
- Demonstrates strong key sensitivity.
- Easy to explain and visualize for learning cryptography concepts.
- Works effectively for small to medium-sized plaintexts.

Weaknesses

- Uses fixed permutation and XOR key, reducing randomness.
- Not scalable for high-security or commercial applications.
- Vulnerable to advanced cryptanalysis if enough ciphertext is available.
- Limited key space compared to modern cryptographic algorithms.

Opportunities

- Can be extended by adding dynamic keys or multiple permutations.

- Useful as a foundation for developing more advanced ciphers.
- Suitable for academic research and student projects.
- Can be enhanced with hashing or authentication mechanisms.
- Useful for teaching cryptography fundamentals in cybersecurity courses.

Threats

- Modern cryptographic algorithms outperform it in security and speed.
- Quantum computing could easily break simple substitution-based schemes.
- Exposure of the algorithm logic reduces security if keys are weak.
- Not suitable for protecting sensitive or classified information.

Application Areas of the Shital Kapari

Education about Cryptography Concepts: Learning how to use methods such as substitution, XOR, permutations and key-based encryption in order to encrypt and decrease file.

Academic Projects: Academic project work and academic study to demonstrate knowledge of the various methods for constructing algorithms.

Communications Secure at Low Levels: To be able to communicate in a secure manner and at low levels of confidentiality.

Obfuscating Non-Critical Data: Ensure that non-critical data cannot be seen the third party.

Cryptography Training Labs: To Learn How to Encrypt or Decrypt Files.

CONCLUSION

This coursework successfully explored the fundamental concepts of cryptography and demonstrated how traditional encryption techniques can be enhanced using mathematical and logical modifications. Beginning with the basic Caesar Cipher, several structured improvements were introduced to design a new cryptographic algorithm known as the Shital Cipher. These modifications significantly increased the complexity and security of the encryption process.

Through the use of reverse alphabet substitution, Caesar shifting with variable keys, ASCII and binary conversions, XOR operations, and permutation techniques, the Shital Cipher provided multiple layers of security. The testing phase confirmed that the algorithm is capable of encrypting and decrypting data accurately while producing highly unpredictable ciphertext. Different test cases, including sensitive data, proved the reliability and consistency of the cipher.

Overall, this coursework demonstrates the practical application of cryptographic system design and highlights how even classical encryption methods can be strengthened to improve data confidentiality. While the Shital Cipher is not intended to replace modern industry-standard algorithms, it effectively illustrates the importance of layered security and serves as a strong educational model for understanding encryption principles in cyber security.

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